

Adani Defence vehicle-mounted counter-drone system (Source: https://bharatshakti.in)

primary mainstays, Navy's HELIOS and the US Air Force's THOR. HELIOS is a 60kW-class system that merges multiple fibre-laser modules into a single, high-quality beam capable of scaling up to 150kW. HELIOS provides three-in-one capability, physically destroying drones via 'hard kills', temporarily blinding optical sensors with 'soft kills', or executing long-range surveillance. THOR uses HPM pulses in a wide conical beam to destroy electronics across hundreds of metres. Other US projects include, The E-HEL, a vehicle-mounted modular laser system scaling up to 50kW for mobile ground forces; the AMP-HEL, a lightweight agile system for rapid, short-range tactical air defence; the HELCAP, a 300kW-class system to intercept fast-moving anti-ship cruise missiles; the SONGBOW, a massive 400kW system combining multiple modules into a single precision beam for defence against hypersonic weapons.

Israel. Israel is leading in active use of DEWs in operations. The Iron Beam, a 100kW fibre laser, integrated into the Iron Dome, a layered national defence network, is capable of intercepting targets at ranges up to 10km. The Iron Beam positions Israel as the first country to successfully deploy HEL weapons in active operations.

China. China is rapidly advancing its DEW arsenal. The combat-proven, 30kW-class 'Silent Hunter' laser provides 'hard kill' against localised targets. The LY-1 (Liaoyuan-1), a 300kW laser, is designed to defend naval vessels and ground forces against low-altitude targets. The truck-mounted Hurricane 3000kW, an HPM system, is being developed 'to fry' targets within a 3km radius. The TPG1000C, a breakthrough gigawatt-level microwave system, is under development to disrupt orbital and upper-atmosphere assets for anti-satellite operations.

The United Kingdom. The UK has operationally deployed DragonFire, an accurate laser system that combines multiple light beams into a single beam making the system stable, accurate, and reliable. On land, DragonFire is mounted on armoured vehicles and naval destroyers for defence from air attacks. Project Icarus is under development as a HPM weapon to fry the electronics of the targets.

Russia. Russia has developed the Peresvet laser system, a high-tech deterrent, as part of space defence strategy to protect strategic nuclear missile bases. Believed to be a massive, megawatt-class weapon, Peresvet tracks and blinds the sensitive optical sensors of spy satellites flying at altitudes of up to 1500km. Despite its huge size and heaviness, the system is fully mobile, mounted on multi-wheeled trucks. Additionally, Russia is focusing on smaller mobile land-based laser systems for the front lines and airborne lasers carried by aircraft and drones.

Other countries. Germany is conducting field-trials of laser-based air defence systems for naval application with power outputs ranging from 20kW to 50kW. France is pursuing a research programme for integrating HELs into air defence networks and making them interoperable with NATO systems. Japan is working on miniaturising compact laser systems for integration into air defence networks. South Korea has a laser weapon development programme aimed at countering UAVs. Turkey has developed ALKA DEW, a mobile tactical system combining laser and electromagnetics for counter-drone missions.

DEWs: Key challenges

DEWs offer innovative strategic systems for integration into defence architecture. However, their deployment has been restricted due to technical restraints, limitations in operations, and ethical requirements that are difficult to comply with. The key constraints include:

Technical constraints. DEWs require massive rapid energy bursts making power management difficult. Generating and storing such huge energy and the associated cooling systems requires large space and adds to weight, restricting deployment on mobile military platforms. Furthermore, achieving desired lethality requires acute tracking precision

Box 4: Sensor fusion

Sensor fusion is an architecture that integrates data from multiple sensors, including radar, infrared, electro-optical cameras, and acoustic detectors into a single coherent operational picture to provide enhanced accuracy, reduced false alarms, and accelerated decision-making. Sensor fusion is a critical enabler for multi-domain operations.